

# Supplementary Information

DrugDis: Disentangling general and context-specific effects in drug-response prediction

This Supplementary Information contains three Supplementary Notes on the response decomposition, zero-amplitude predictions and cross-assay reproducibility; 22 Supplementary Tables with detailed benchmark results; and one Supplementary Data item describing the data sources.

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## Supplementary Notes

### Note 1: the decomposition on an unbalanced observed support

The projection  $H$  is fixed by the observed support alone, and the support of a pharmacogenomic dataset is incomplete and unbalanced. Three properties of the decomposition follow.

**The drug and sample marginals share part of the additive variance.** On a complete balanced grid the two sets of indicator columns are orthogonal, and the additive variance splits uniquely into a drug part and a sample part. On an unbalanced support they are correlated, so the additive variance divides into a part unique to each marginal and a shared part. On the benchmark dataset the drug marginal alone accounts for 72.65% of total response variance and the sample marginal alone for 4.17%, and 0.97% is shared. The unique parts are 71.68% and 3.20%, and the three sum to the 75.85% the additive component carries. Across the ten test sets the shared part ranges from 0.46% to 1.40% (Supplementary Table 2). The main text quotes the unique shares.

**The interaction component is indexed by the support that defines it.** Two evaluation sets with different supports define different projections, so each defines its own interaction component. The dimension of the additive subspace is  $\text{rank}(H) = n_{\text{drug}} + n_{\text{sample}} - c$  with  $c$  the number of connected components of the bipartite support graph, and  $c = 1$  for the benchmark dataset and for every test set. The fraction of degrees of freedom the projection removes varies by an order of magnitude between evaluation sets: 1.76% on the benchmark dataset, 9.52% to 11.45% on the held-out cell-line test sets, and 1.92% to 1.98% on the held-out compound test sets. The error density in Methods is therefore normalized by the rank of each subspace, and component values are compared within a distribution-shift regime. A compound or sample observed once has its response as its own marginal, so the projection assigns that response entirely to the additive component; sparsely observed compounds and samples contribute to the additive component by construction.

**The marginals reflect the screening design as well as the biology.** The observed support covers 3,141,680 of the  $54,180 \times 986$  possible drug-sample pairs, 5.88% of the grid, and the screened pairs were chosen by the resources that ran them. The drug marginal is the mean response of a compound over the samples on which it was screened, and the sample marginal is the mean response of a sample over the compounds applied to it. The additive component is therefore defined by the observed design as well as by pharmacology, and the interaction component is the deviation from the best additive fit to that design. Every component value in this work is stated on a named support: the benchmark dataset, a test set or a single response resource (Supplementary Table 3).

### Note 2: the degenerate case of the interaction-component correlation

The interaction-component correlation is  $\text{corr}(My, M\hat{y})$ . When the prediction lies in the additive subspace,  $M\hat{y} = 0$ , both the numerator and the denominator of the Pearson formula vanish, and the quantity is undefined. This is the situation of every predictor that carries no drug-sample interaction information.

**Which predictors fall in it.** A constant predictor does, and so does any predictor assembled from drug-level and sample-level terms, including one built from training-set marginals. A structurally additive decoder does as well, for the same reason. Across the 28 runs of the additive ridge decoder in the decoder comparison, the standard deviation of the emitted interaction lies between  $1.6 \times 10^{-9}$  and  $1.2 \times 10^{-8}$ , against an observed interaction standard deviation of order  $7 \times 10^{-2}$ , and the emitted amplitude  $A_{\text{int}}$  is zero to six decimal places in every one (Supplementary Table 11). The Pearson formula returns  $-0.019$  to  $+0.017$  for those runs, the correlation of floating-point noise, so the main text compares these decoders by emitted amplitude (Fig. 3g).

**How the two cases are separated.** We record the emitted amplitude  $A_{\text{int}} = \text{sd}(M\hat{y})/\text{sd}(My)$  beside every correlation. A prediction is classed as emitting no interaction when  $A_{\text{int}} \leq 10^{-8}$ , and a decoder that is additive by construction is classed as non-emitting whatever its finite-precision residual, as for the additive ridge in Fig. 3g. In the null calibration the threshold falls in an empty region: predictors

that emit nothing reach  $9.4 \times 10^{-18}$  to  $1.8 \times 10^{-9}$ , whereas emitting predictions range from 0.2775 to 1.9275. For a non-emitting prediction the interaction-component correlation is reported as undefined, as in Supplementary Table 5. A constant predictor carries no variance in any subspace, so all three of its correlations are undefined.

**Shuffled predictions set the null scale.** A predictor that emits interaction but carries none of its structure gives a measured zero on a defined quantity. Shuffled predictions emit 1.3480 to 1.9275 times the amplitude the observed interaction contains and score  $-0.001$  under held-out cell lines,  $+0.000$  under held-out compounds and  $+0.000$  in organoids, which calibrates the scale of the interaction-component correlation.

### Note 3: interpretation of the cross-assay reproducibility reference

Cross-assay reproducibility  $q_c$  is the correlation between GDSC1 and GDSC2 measurements of the same drug-sample pair, computed component by component with one projection built on the matched support and applied identically to both arms. On the 32,724 matched pairs that meet the benchmark eligibility rules it is 0.650 for total response, 0.754 for the additive component and 0.498 for the interaction component. The empirical reproducibility reference is  $\sqrt{q_c}$ , that is 0.807, 0.869 and 0.706, and recovery relative to it is  $\eta_c = r_c / \sqrt{q_c}$  (Supplementary Table 6).

The square root follows from a classical measurement model. If two measurements obey  $y_A = z + \epsilon_A$  and  $y_B = z + \epsilon_B$ , where  $z$  is a common latent quantity, the errors have equal variance, are mutually independent and are independent of  $z$ , then  $q_c = \text{Var}(z) / \text{Var}(y_A)$  and a perfect predictor of  $z$  has correlation  $\sqrt{q_c}$  with one measured arm. GDSC1 and GDSC2 differ in assay generation, so their errors need not be exchangeable or independent, and  $\sqrt{q_c}$  serves as an empirical comparison reference.

**Analysis supports.** The analyses use the following subsets of matched observations, all on DROMA Sensitivity without rescaling.

| Stage                               | Pairs and transformation                                                                   | Count                                                                                     | Use                                                                                                                  |
|-------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Benchmark response panel            | Canonical SMILES and DROMA sample identifier; mean of the retained measurements            | 3,141,680                                                                                 | Training and evaluation target for the benchmark.                                                                    |
| GDSC1-GDSC2 match                   | Exactly one GDSC1 and one GDSC2 measurement for the same canonical pair, giving $y_A, y_B$ | 43,622; 89 drugs, 799 samples                                                             | Starting set for cross-assay reproducibility and independent model-selection analyses.                               |
| Benchmark eligibility filter        | Matched pairs that meet the benchmark eligibility rules                                    | 32,724; 89 drugs, 600 samples                                                             | Full-support $q_c$ reference.                                                                                        |
| Exact recovery support              | Matched test pairs whose benchmark target equals $y_A$ within $10^{-6}$                    | Held-out cell lines 2,876-3,051; held-out compounds 1,970-3,873 per seed                  | Support for $\eta_c$ ; the target-equality filter excludes pairs whose benchmark target comes from another resource. |
| Independent model-selection support | Matched pairs in the test set of each split and seed, shared by M0 and M2                  | Held-out cell lines 4,927-5,411 (mean 5,138); held-out compounds 2,723-7,164 (mean 5,027) | GDSC1 selection metrics and GDSC2 independent endpoint.                                                              |

The reference has four properties.

**It measures reproducibility across assay generations.** The two arms are GDSC1 and GDSC2, two assay generations run by one consortium, so the estimate combines measurement noise with the assay change. Pairings across different resources span 389 to 3,174 pairs on fewer than 20 compounds and add protocol divergence to the same assay difference; GDSC1-GDSC2 was therefore adopted as the reference.

**It summarizes a magnitude-dependent reproducibility.** On the matched pairs,  $q_{\text{int}}$  rises from 0.08 in the smallest quartile of interaction magnitude to 0.62 in the largest, and one scalar per component summarizes that range. Recovery ratios can therefore exceed one: under held-out cell lines the additive  $\eta$  averages 1.00 across five seeds, with a range of 0.97 to 1.03 (Supplementary Table 14).

**It is defined where measurements repeat.** On the pairs each regime holds out, the support is 2,876 to 3,051 pairs under held-out cell lines, over 79 to 86 compounds and 90 to 101 held-out cell lines. Under held-out compounds it is 1,970 to 3,873 pairs on 7 to 19 held-out compounds; across the five seeds the ratio then ranges from 0.03 to 0.54 for the interaction component and from 0.32 to 0.65 for the additive component, and the per-seed intervals for total response and for the additive component include zero at three or more seeds in either training arm.  $\eta_c$  is therefore reported under held-out cell lines, and Supplementary Table 14 lists the per-seed values in both regimes. The organoid cohorts carry no repeated cross-assay measurements.

**Its uncertainty is dominated by the compound axis.** Observations sharing a compound or a sample are correlated, so intervals come from cluster bootstraps. For total response the sample bootstrap returns 0.640 to 0.660 across 600 clusters, while the compound bootstrap returns 0.529 to 0.755 across 89. The two-way scheme, which resamples both axes with draw multiplicity retained, gives 0.528 to 0.750. Intervals on  $\eta_c$  resample both axes once per replicate and recompute the model correlation and the reference together, so the numerator and the denominator carry their uncertainty jointly.

On the exact recovery support the scored target equals the single arm-A measurement, so the reference applies without the Spearman–Brown rescaling  $q_n = nq / (1 + (n - 1)q)$  that an average of  $n$  measurements would require (Spearman, *Br. J. Psychol.* **3**, 271–295, 1910; Brown, *Br. J. Psychol.* **3**, 296–322, 1910).

## Supplementary Tables

**Supplementary Table 1: Benchmark dataset composition.** The transcriptomic input matrix is CCLE-derived, restricted to 15,961 genes and holds one profile per sample. A sample enters the benchmark dataset only if a CCLE baseline profile exists for it. Response rows are counted per resource after the overlap rule of the processed response table and before rows sharing a canonical drug-sample pair were averaged (Methods).

| Item                                       | Value              |
|--------------------------------------------|--------------------|
| Transcriptomic input                       | CCLE, 15,961 genes |
| Profiled samples in the input matrix       | 1,406              |
| Drug-sample pairs in the benchmark dataset | 3,141,680          |
| Cell-line samples                          | 986                |
| Compounds                                  | 54,180             |
| NCI60 (response rows)                      | 2,175,581          |
| PRISM (response rows)                      | 504,493            |
| CTRP2 (response rows)                      | 269,725            |
| GDSC1 (response rows)                      | 125,914            |
| GDSC2 (response rows)                      | 44,797             |
| CTRP1 (response rows)                      | 39,784             |
| CCLE (response rows)                       | 10,730             |
| GRAY (response rows)                       | 1,888              |
| gCSI (response rows)                       | 1,646              |
| FIMM (response rows)                       | 890                |
| UHNBreast (response rows)                  | 54                 |
| <b>All response resources</b>              | <b>3,175,502</b>   |

**Supplementary Table 2: Orthogonal response decomposition on the benchmark dataset and on every test set.**  $\text{rank}(H)$  is the dimension of the additive subspace and  $d_H/N$  the fraction of degrees of freedom it occupies, in per cent. Additive and Interaction are the percentages of total response variance carried by the two components. Drug and Sample are the percentages uniquely attributable to each marginal and Shared the percentage carried by both (Supplementary Note 1).  $|\text{corr}|$  is the absolute correlation between the two components, in units of  $10^{-11}$ .

| Evaluation set      | Seed | Pairs     | Compounds | Samples | $\text{rank}(H)$ | $d_H/N$ | Additive | Interaction | Drug  | Sample | Shared | $ \text{corr} $ |
|---------------------|------|-----------|-----------|---------|------------------|---------|----------|-------------|-------|--------|--------|-----------------|
| Benchmark dataset   | —    | 3,141,680 | 54,180    | 986     | 55,165           | 1.76    | 75.85    | 24.15       | 71.68 | 3.20   | 0.97   | 2.72            |
| Held-out cell lines | 3407 | 557,484   | 54,176    | 148     | 54,323           | 9.74    | 78.36    | 21.64       | 74.62 | 2.34   | 1.40   | 1.39            |
|                     | 3408 | 474,279   | 54,176    | 148     | 54,323           | 11.45   | 76.22    | 23.78       | 71.31 | 4.46   | 0.46   | 1.68            |
|                     | 3409 | 486,991   | 54,172    | 148     | 54,319           | 11.15   | 77.16    | 22.84       | 73.81 | 2.46   | 0.90   | 3.20            |
|                     | 3410 | 570,863   | 54,175    | 148     | 54,322           | 9.52    | 76.76    | 23.24       | 72.24 | 3.86   | 0.67   | 2.58            |
|                     | 3411 | 530,351   | 54,175    | 148     | 54,322           | 10.24   | 76.93    | 23.07       | 73.40 | 2.78   | 0.75   | 2.68            |
| Held-out compounds  | 3407 | 468,871   | 8,127     | 986     | 9,112            | 1.94    | 74.78    | 25.22       | 70.73 | 3.19   | 0.86   | 2.09            |
|                     | 3408 | 466,230   | 8,127     | 984     | 9,110            | 1.95    | 75.83    | 24.17       | 71.82 | 2.98   | 1.03   | 2.04            |
|                     | 3409 | 459,058   | 8,127     | 984     | 9,110            | 1.98    | 75.93    | 24.07       | 71.56 | 3.37   | 1.00   | 2.62            |
|                     | 3410 | 475,001   | 8,127     | 986     | 9,112            | 1.92    | 75.82    | 24.18       | 71.64 | 3.31   | 0.87   | 1.01            |
|                     | 3411 | 474,086   | 8,127     | 984     | 9,110            | 1.92    | 76.17    | 23.83       | 72.45 | 3.19   | 0.53   | 0.57            |

**Supplementary Table 3: Orthogonal response decomposition within each response resource.** Each resource is decomposed on its own eligible screen: its DROMA measurements that meet the inclusion criteria of the benchmark dataset, with rows sharing a canonical drug-sample pair averaged within the resource and no value from another resource. The benchmark dataset row repeats Supplementary Table 2. Without NCI60 is the benchmark dataset rebuilt from the other ten resources by the rules that built it. Resources are ordered by the number of pairs in their screen. Columns as in Supplementary Table 2, in per cent. Shared is the inclusion-exclusion overlap of the two marginal subspaces, which are correlated on an unbalanced support, and can be slightly negative (Supplementary Note 1). <sup>b</sup>Pairs and additive share on the rows that the benchmark dataset attributes to the resource, which are fewer where the same compound name and sample were also measured by a resource earlier in the order of the overlap rule (Methods). The absolute correlation between the two components is below  $10^{-9}$  in every row.

| Response support  | Pairs     | Compounds | Samples | $d_H/N$ | Additive | Interaction | Drug  | Sample | Shared | Pairs <sup>b</sup> | Additive <sup>b</sup> |
|-------------------|-----------|-----------|---------|---------|----------|-------------|-------|--------|--------|--------------------|-----------------------|
| Benchmark dataset | 3,141,680 | 54,180    | 986     | 1.76    | 75.85    | 24.15       | 71.68 | 3.20   | 0.97   | —                  | —                     |
| Without NCI60     | 993,482   | 1,926     | 984     | 0.29    | 66.54    | 33.46       | 60.59 | 4.13   | 1.83   | —                  | —                     |
| NCI60             | 2,170,974 | 52,908    | 68      | 2.44    | 82.54    | 17.46       | 79.48 | 2.69   | 0.37   | 2,157,687          | 82.59                 |
| PRISM             | 612,596   | 1,375     | 473     | 0.30    | 81.27    | 18.73       | 79.37 | 1.90   | -0.01  | 503,644            | 81.61                 |
| CTRP2             | 292,817   | 462       | 822     | 0.44    | 69.74    | 30.26       | 60.06 | 8.55   | 1.13   | 269,725            | 69.94                 |
| GDSC1             | 167,886   | 293       | 701     | 0.59    | 59.04    | 40.96       | 53.17 | 5.93   | -0.06  | 125,914            | 58.17                 |
| GDSC2             | 88,507    | 174       | 605     | 0.88    | 59.03    | 40.97       | 52.37 | 6.66   | 0.00   | 44,797             | 56.01                 |
| CTRP1             | 40,085    | 307       | 234     | 1.35    | 73.17    | 26.83       | 57.82 | 14.04  | 1.31   | 39,784             | 73.32                 |
| gCSI              | 11,495    | 41        | 523     | 4.90    | 71.51    | 28.49       | 65.13 | 7.31   | -0.93  | 1,646              | 83.50                 |
| CCLC              | 10,730    | 24        | 472     | 4.61    | 79.27    | 20.73       | 73.99 | 5.02   | 0.27   | 10,730             | 79.27                 |
| GRAY              | 3,087     | 95        | 50      | 4.66    | 72.75    | 27.25       | 65.04 | 7.84   | -0.13  | 1,888              | 74.76                 |
| FIMM              | 2,345     | 52        | 46      | 4.14    | 56.59    | 43.41       | 46.73 | 9.80   | 0.06   | 890                | 58.13                 |
| UHNBreast         | 166       | 6         | 39      | 26.51   | 48.04    | 51.96       | 24.45 | 21.55  | 2.04   | 54                 | 87.98                 |

**Supplementary Table 4: Behaviour of existing evaluation views and DrugDis under prediction perturbations.** Entries are mean changes from the original prediction over M0 and M2 at five seeds. Each metric is computed on identical predictions and pair support within a run. Drug-only and cell-only offsets have a standard deviation equal to half that of the observed response. The two interaction perturbations retain the projected additive prediction and multiply only the projected interaction prediction.

| Regime              | Metric                          | Drug-only offset | Cell-only offset | Interaction $\times 0.5$ | Interaction $\times 2$ |
|---------------------|---------------------------------|------------------|------------------|--------------------------|------------------------|
| Held-out cell lines | Global                          | -0.1185          | -0.1216          | -0.0015                  | -0.0209                |
|                     | Fixed drug                      | 0.0000           | -0.1203          | 0.0111                   | -0.0471                |
|                     | Fixed cell                      | -0.1056          | 0.0000           | -0.0014                  | -0.0324                |
|                     | DrEval normalized               | -0.1544          | -0.1663          | -0.0485                  | 0.0230                 |
|                     | DrugDis interaction correlation | -0.0000          | 0.0000           | 0.0000                   | 0.0000                 |
|                     | DrugDis interaction amplitude   | 0.0000           | -0.0000          | -0.1807                  | 0.3614                 |
|                     | DrugDis interaction $R^2$       | -0.0000          | 0.0000           | -0.0138                  | -0.1702                |
| Held-out compounds  | Global                          | -0.1230          | -0.1207          | -0.0030                  | -0.0216                |
|                     | Fixed drug                      | 0.0000           | -0.2051          | 0.0021                   | -0.0950                |
|                     | Fixed cell                      | -0.1111          | -0.0000          | 0.0008                   | -0.0291                |
|                     | DrEval normalized               | -0.1254          | -0.1232          | -0.0035                  | -0.0213                |
|                     | DrugDis interaction correlation | 0.0000           | 0.0000           | 0.0000                   | 0.0000                 |
|                     | DrugDis interaction amplitude   | 0.0000           | -0.0000          | -0.1833                  | 0.3667                 |
|                     | DrugDis interaction $R^2$       | 0.0000           | 0.0000           | -0.0088                  | -0.1842                |

**Supplementary Table 5: Null-predictor calibration.** Mean and standard deviation across five seeds for four predictors under each evaluation regime. Total, Additive and Interaction are the three component-wise correlations. Entries read *undefined* where the predicted vector carries no variance in the corresponding subspace: for the constant predictor in all three, and for the constant and marginal predictors in the interaction subspace, which is the case Supplementary Note 2 treats.  $\text{sd}(M\hat{y})$  is the emitted interaction amplitude in absolute units and  $\text{sd}(My)$  the amplitude the observed interaction carries. The organoid rows are the primary pooled set (UMPDO1–3).

| Regime                    | Predictor              | $n$ | Total          | Additive       | Interaction    | $\text{sd}(M\hat{y})$  | $\text{sd}(My)$ |
|---------------------------|------------------------|-----|----------------|----------------|----------------|------------------------|-----------------|
| Held-out cell lines       | Constant               | 5   | undefined      | undefined      | undefined      | $1.88 \times 10^{-17}$ | 0.0711          |
|                           | Training marginals     | 5   | 0.849 (0.010)  | 0.967 (0.007)  | undefined      | $1.29 \times 10^{-18}$ | 0.0711          |
|                           | Shuffled predictions   | 5   | -0.001 (0.001) | -0.001 (0.005) | -0.001 (0.002) | 0.1203                 | 0.0711          |
|                           | Trained predictor (M0) | 5   | 0.858 (0.009)  | 0.963 (0.007)  | 0.315 (0.040)  | 0.0247                 | 0.0711          |
| Held-out compounds        | Constant               | 5   | undefined      | undefined      | undefined      | $1.25 \times 10^{-16}$ | 0.0732          |
|                           | Training marginals     | 5   | 0.196 (0.006)  | 0.226 (0.008)  | undefined      | $1.22 \times 10^{-10}$ | 0.0732          |
|                           | Shuffled predictions   | 5   | 0.000 (0.002)  | -0.002 (0.011) | 0.000 (0.001)  | 0.1037                 | 0.0732          |
|                           | Trained predictor (M0) | 5   | 0.623 (0.008)  | 0.697 (0.006)  | 0.291 (0.017)  | 0.0274                 | 0.0732          |
| Organoids, primary pooled | Constant               | 5   | undefined      | undefined      | undefined      | $8.72 \times 10^{-18}$ | 0.2407          |
|                           | Training marginals     | 5   | 0.465 (0.001)  | 0.588 (0.001)  | undefined      | $6.59 \times 10^{-18}$ | 0.2407          |
|                           | Shuffled predictions   | 5   | 0.007 (0.015)  | 0.050 (0.065)  | 0.000 (0.013)  | 0.3878                 | 0.2407          |

**Supplementary Table 6: Cross-assay reproducibility of each response component, with cluster-bootstrap intervals.**  $q_c$  is the correlation between the GDSC1 and GDSC2 measurements of the same drug-sample pair, estimated on the 32,724 matched pairs that meet the benchmark eligibility rules, over 89 compounds and 600 samples. Each bootstrap column gives the replicate mean and the 95% interval under one resampling scheme, 600 replicates each; the decomposition is rebuilt inside every replicate.  $\sqrt{q_c}$  is the empirical reproducibility reference against which recovery is expressed; Supplementary Note 3 gives its assumptions.

| Component             | $q_c$ | Two-way cluster      | Compound cluster     | Sample cluster       | $\sqrt{q_c}$ |
|-----------------------|-------|----------------------|----------------------|----------------------|--------------|
| Total response        | 0.650 | 0.636 [0.528, 0.750] | 0.636 [0.529, 0.755] | 0.650 [0.640, 0.660] | 0.807        |
| Additive component    | 0.754 | 0.728 [0.567, 0.866] | 0.728 [0.562, 0.868] | 0.754 [0.743, 0.764] | 0.869        |
| Interaction component | 0.498 | 0.498 [0.444, 0.554] | 0.499 [0.451, 0.547] | 0.498 [0.481, 0.515] | 0.706        |

**Supplementary Table 7: Resource-composition sensitivity of the M2–M0 interaction effect.** Differences are paired within seed. Positive gives the number of seeds whose interaction-correlation difference is positive. Each support contributes at least 1,000 test pairs and is a rescored subset of the same trained predictions.

| Regime              | Support          | Mean pairs | Seeds | Mean $\Delta r_{\text{int}}$ | s.d.   | Positive | Mean $\Delta R_{\text{int}}^2$ |
|---------------------|------------------|------------|-------|------------------------------|--------|----------|--------------------------------|
| Held-out cell lines | NCI60            | 369,208    | 5     | -0.0220                      | 0.0368 | 1/5      | -0.0139                        |
|                     | All except NCI60 | 155,147    | 5     | 0.0078                       | 0.0227 | 4/5      | 0.0020                         |
|                     | PRISM            | 80,984     | 5     | 0.0240                       | 0.0243 | 5/5      | 0.0223                         |
|                     | GDSC1            | 19,096     | 5     | -0.0004                      | 0.0070 | 2/5      | -0.0039                        |
|                     | GDSC2            | 6,971      | 5     | -0.0083                      | 0.0406 | 3/5      | -0.0229                        |
|                     | CTRP2            | 42,081     | 5     | -0.0302                      | 0.0514 | 1/5      | -0.0466                        |
|                     | CTRP1            | 5,766      | 5     | 0.0090                       | 0.0387 | 3/5      | -0.0069                        |
|                     | CACLE            | 1,700      | 5     | -0.0024                      | 0.0233 | 2/5      | -0.0047                        |
| Held-out compounds  | NCI60            | 323,805    | 5     | 0.0237                       | 0.0170 | 5/5      | 0.0216                         |
|                     | All except NCI60 | 145,156    | 5     | 0.0081                       | 0.0039 | 5/5      | 0.0110                         |
|                     | PRISM            | 72,812     | 5     | 0.0181                       | 0.0041 | 5/5      | 0.0264                         |
|                     | GDSC1            | 16,767     | 5     | 0.0040                       | 0.0228 | 2/5      | 0.0102                         |
|                     | GDSC2            | 7,093      | 5     | -0.0127                      | 0.0310 | 1/5      | 0.0250                         |
|                     | CTRP2            | 40,817     | 5     | 0.0247                       | 0.0251 | 4/5      | 0.0187                         |
|                     | CTRP1            | 6,111      | 5     | 0.0105                       | 0.0200 | 3/5      | 0.0110                         |
|                     | CACLE            | 2,539      | 4     | 0.0137                       | 0.0624 | 3/4      | 0.0121                         |



**Supplementary Table 8: Representation benchmark under held-out cell lines.** Mean and standard deviation across seeds for every configuration, under a fixed nonlinear dual-tower decoder. *Full panel* records whether the representation covers the entire benchmark dataset; the two that do not are scored on the subset they cover.  $A_{\text{int}}$  is the emitted interaction amplitude. The model arms M0 and M2 are listed first and carry five seeds where the representations carry three. Expression is the untransformed gene-level expression input.

| Class                          | Representation                 | Full panel | Seeds | Total         | Additive      | Interaction   | $A_{\text{int}}$ |
|--------------------------------|--------------------------------|------------|-------|---------------|---------------|---------------|------------------|
| Model arms                     | M0 (ECFP4 $\times$ Expression) | yes        | 5     | 0.858 (0.009) | 0.963 (0.007) | 0.315 (0.040) | 0.346 (0.042)    |
|                                | M2 (ECFP4 $\times$ Expression) | yes        | 5     | 0.858 (0.006) | 0.963 (0.004) | 0.307 (0.042) | 0.376 (0.028)    |
| Molecular representations      | ECFP4                          | yes        | 3     | 0.859 (0.007) | 0.964 (0.003) | 0.308 (0.053) | 0.356 (0.010)    |
|                                | GeminiMol                      | no         | 3     | 0.828 (0.010) | 0.933 (0.007) | 0.283 (0.047) | 0.323 (0.030)    |
|                                | Mole_BERT                      | yes        | 3     | 0.827 (0.008) | 0.929 (0.004) | 0.274 (0.039) | 0.338 (0.053)    |
|                                | ChemBERTa2                     | yes        | 3     | 0.821 (0.011) | 0.925 (0.007) | 0.255 (0.057) | 0.328 (0.014)    |
|                                | MolT5                          | yes        | 3     | 0.829 (0.012) | 0.934 (0.007) | 0.249 (0.078) | 0.324 (0.025)    |
|                                | MolCLR                         | yes        | 3     | 0.742 (0.013) | 0.834 (0.008) | 0.238 (0.073) | 0.231 (0.017)    |
|                                | KPGT                           | yes        | 3     | 0.769 (0.016) | 0.865 (0.019) | 0.236 (0.069) | 0.286 (0.037)    |
|                                | UniMolV2                       | no         | 3     | 0.757 (0.028) | 0.853 (0.029) | 0.225 (0.035) | 0.276 (0.039)    |
|                                | UniMol                         | yes        | 3     | 0.768 (0.020) | 0.868 (0.015) | 0.208 (0.024) | 0.292 (0.049)    |
|                                | InfoAlign                      | yes        | 3     | 0.757 (0.022) | 0.862 (0.015) | 0.183 (0.106) | 0.385 (0.031)    |
|                                | Ouroboros                      | yes        | 3     | 0.678 (0.018) | 0.766 (0.016) | 0.170 (0.030) | 0.222 (0.016)    |
|                                | Chemprop                       | yes        | 3     | 0.513 (0.021) | 0.587 (0.031) | 0.118 (0.002) | 0.284 (0.047)    |
| Transcriptomic representations | SCimilarity                    | yes        | 3     | 0.865 (0.004) | 0.969 (0.005) | 0.318 (0.054) | 0.350 (0.035)    |
|                                | Expression                     | yes        | 3     | 0.859 (0.007) | 0.964 (0.003) | 0.308 (0.053) | 0.356 (0.010)    |
|                                | Tahoe_x1                       | yes        | 3     | 0.857 (0.007) | 0.963 (0.004) | 0.293 (0.050) | 0.368 (0.029)    |
|                                | scGPT                          | yes        | 3     | 0.860 (0.006) | 0.967 (0.003) | 0.290 (0.019) | 0.374 (0.022)    |
|                                | STATE                          | yes        | 3     | 0.849 (0.014) | 0.959 (0.007) | 0.276 (0.073) | 0.433 (0.011)    |
|                                | scFoundation                   | yes        | 3     | 0.858 (0.003) | 0.967 (0.002) | 0.272 (0.064) | 0.376 (0.074)    |
|                                | CellPLM                        | yes        | 3     | 0.857 (0.013) | 0.964 (0.008) | 0.271 (0.053) | 0.306 (0.042)    |
|                                | Cell2Sentence                  | yes        | 3     | 0.852 (0.015) | 0.963 (0.012) | 0.257 (0.034) | 0.420 (0.035)    |
|                                | BulkFormer                     | yes        | 3     | 0.849 (0.017) | 0.960 (0.009) | 0.252 (0.030) | 0.356 (0.085)    |
|                                | Geneformer                     | yes        | 3     | 0.853 (0.012) | 0.965 (0.011) | 0.243 (0.056) | 0.392 (0.024)    |
|                                | tGPT                           | yes        | 3     | 0.846 (0.015) | 0.961 (0.009) | 0.209 (0.020) | 0.391 (0.017)    |
|                                | UCE                            | yes        | 3     | 0.813 (0.006) | 0.938 (0.015) | 0.013 (0.049) | 0.288 (0.069)    |

**Supplementary Table 9: Representation benchmark under held-out compounds.** Columns as in Supplementary Table 8.

| Class                          | Representation                 | Full panel | Seeds | Total         | Additive      | Interaction    | $A_{\text{int}}$ |
|--------------------------------|--------------------------------|------------|-------|---------------|---------------|----------------|------------------|
| Model arms                     | M2 (ECFP4 $\times$ Expression) | yes        | 5     | 0.609 (0.013) | 0.678 (0.014) | 0.308 (0.013)  | 0.359 (0.012)    |
|                                | M0 (ECFP4 $\times$ Expression) | yes        | 5     | 0.623 (0.008) | 0.697 (0.006) | 0.291 (0.017)  | 0.374 (0.007)    |
| Molecular representations      | ECFP4                          | yes        | 3     | 0.625 (0.010) | 0.699 (0.008) | 0.301 (0.014)  | 0.371 (0.002)    |
|                                | GeminiMol                      | no         | 3     | 0.585 (0.016) | 0.663 (0.016) | 0.232 (0.012)  | 0.313 (0.033)    |
|                                | Mole_BERT                      | yes        | 3     | 0.575 (0.017) | 0.649 (0.016) | 0.228 (0.011)  | 0.327 (0.025)    |
|                                | MolCLR                         | yes        | 3     | 0.573 (0.009) | 0.648 (0.011) | 0.224 (0.017)  | 0.288 (0.007)    |
|                                | MolT5                          | yes        | 3     | 0.531 (0.005) | 0.599 (0.004) | 0.214 (0.008)  | 0.296 (0.017)    |
|                                | UniMolV2                       | no         | 3     | 0.565 (0.012) | 0.640 (0.013) | 0.207 (0.005)  | 0.280 (0.009)    |
|                                | UniMol                         | yes        | 3     | 0.520 (0.019) | 0.590 (0.020) | 0.189 (0.003)  | 0.302 (0.013)    |
|                                | Ouroboros                      | yes        | 3     | 0.537 (0.007) | 0.611 (0.005) | 0.186 (0.003)  | 0.302 (0.026)    |
|                                | ChemBERTa2                     | yes        | 3     | 0.512 (0.003) | 0.581 (0.006) | 0.183 (0.005)  | 0.316 (0.038)    |
|                                | KPGT                           | yes        | 3     | 0.360 (0.021) | 0.411 (0.024) | 0.097 (0.011)  | 0.186 (0.043)    |
|                                | Chemprop                       | yes        | 3     | 0.173 (0.014) | 0.201 (0.015) | 0.006 (0.011)  | 0.058 (0.011)    |
|                                | InfoAlign                      | yes        | 3     | 0.160 (0.007) | 0.187 (0.007) | -0.001 (0.005) | 0.067 (0.004)    |
| Transcriptomic representations | STATE                          | yes        | 3     | 0.628 (0.009) | 0.702 (0.007) | 0.303 (0.019)  | 0.400 (0.019)    |
|                                | scGPT                          | yes        | 3     | 0.625 (0.012) | 0.698 (0.012) | 0.303 (0.013)  | 0.378 (0.009)    |
|                                | Expression                     | yes        | 3     | 0.625 (0.010) | 0.699 (0.008) | 0.301 (0.014)  | 0.371 (0.002)    |
|                                | Tahoe_x1                       | yes        | 3     | 0.623 (0.008) | 0.697 (0.008) | 0.300 (0.003)  | 0.384 (0.033)    |
|                                | scFoundation                   | yes        | 3     | 0.620 (0.007) | 0.693 (0.007) | 0.296 (0.009)  | 0.376 (0.022)    |
|                                | BulkFormer                     | yes        | 3     | 0.620 (0.017) | 0.694 (0.016) | 0.293 (0.012)  | 0.384 (0.015)    |
|                                | SCimilarity                    | yes        | 3     | 0.613 (0.012) | 0.686 (0.013) | 0.290 (0.006)  | 0.353 (0.021)    |
|                                | tGPT                           | yes        | 3     | 0.615 (0.016) | 0.689 (0.017) | 0.287 (0.016)  | 0.376 (0.023)    |
|                                | CellPLM                        | yes        | 3     | 0.614 (0.016) | 0.687 (0.015) | 0.285 (0.017)  | 0.364 (0.024)    |
|                                | Cell2Sentence                  | yes        | 3     | 0.618 (0.009) | 0.693 (0.007) | 0.280 (0.011)  | 0.342 (0.018)    |
|                                | UCE                            | yes        | 3     | 0.618 (0.006) | 0.693 (0.006) | 0.276 (0.011)  | 0.355 (0.030)    |
|                                | Geneformer                     | yes        | 3     | 0.610 (0.011) | 0.685 (0.011) | 0.268 (0.013)  | 0.339 (0.020)    |



**Supplementary Table 10: Descriptive score separation in the representation benchmark.** Spread is the range of the mean score across representations on one axis, median seed s.d. is the median across representations of the seed-to-seed standard deviation, and Ratio is their quotient, which describes score separation relative to seed variation.

| Regime              | Axis           | Component             | Spread | Median seed s.d. | Ratio |
|---------------------|----------------|-----------------------|--------|------------------|-------|
| Held-out cell lines | Molecular      | Total response        | 0.3464 | 0.0147           | 23.6  |
|                     |                | Additive component    | 0.3771 | 0.0114           | 33.0  |
|                     |                | Interaction component | 0.1897 | 0.0550           | 3.5   |
|                     | Transcriptomic | Total response        | 0.0520 | 0.0097           | 5.4   |
|                     |                | Additive component    | 0.0310 | 0.0079           | 3.9   |
|                     |                | Interaction component | 0.3055 | 0.0514           | 5.9   |
| Held-out compounds  | Molecular      | Total response        | 0.4650 | 0.0097           | 47.9  |
|                     |                | Additive component    | 0.5119 | 0.0095           | 54.0  |
|                     |                | Interaction component | 0.3020 | 0.0095           | 31.9  |
|                     | Transcriptomic | Total response        | 0.0186 | 0.0106           | 1.8   |
|                     |                | Additive component    | 0.0175 | 0.0098           | 1.8   |
|                     |                | Interaction component | 0.0352 | 0.0126           | 2.8   |

**Supplementary Table 11: Component-wise recovery under three decoders.** Mean and standard deviation across the runs of each configuration. The additive ridge decoder is additive by construction, so its prediction lies in the additive subspace and it emits an interaction of standard deviation  $10^{-9}$  to  $10^{-8}$ , against an observed interaction standard deviation of order  $7 \times 10^{-2}$ ; its interaction-component correlations are correlations of floating-point noise (Supplementary Note 2). Three of the sixty low-rank bilinear runs, all at the largest weight decay, collapse to the same condition.  $sd(M\hat{y})$  is the emitted interaction amplitude in absolute units.

| Decoder              | Molecular | Transcriptomic | Runs | Parameters | Total         | Additive      | Interaction    | $sd(M\hat{y})$ |
|----------------------|-----------|----------------|------|------------|---------------|---------------|----------------|----------------|
| Additive ridge       | ECFP4     | BulkFormer     | 4    | 2,689      | 0.408 (0.103) | 0.461 (0.117) | 0.009 (0.007)  | 0.0000         |
|                      | ECFP4     | CellPLM        | 4    | 2,561      | 0.483 (0.111) | 0.546 (0.125) | 0.005 (0.004)  | 0.0000         |
|                      | ECFP4     | Expression     | 4    | 18,010     | 0.254 (0.183) | 0.287 (0.206) | 0.004 (0.006)  | 0.0000         |
|                      | ECFP4     | scGPT          | 4    | 2,561      | 0.470 (0.094) | 0.531 (0.106) | -0.007 (0.010) | 0.0000         |
|                      | KPGT      | Expression     | 4    | 18,266     | 0.176 (0.067) | 0.199 (0.076) | -0.001 (0.002) | 0.0000         |
|                      | MolCLR    | Expression     | 4    | 16,474     | 0.099 (0.094) | 0.112 (0.106) | -0.000 (0.001) | 0.0000         |
|                      | UniMol    | Expression     | 4    | 16,474     | 0.151 (0.124) | 0.171 (0.140) | 0.000 (0.001)  | 0.0000         |
| Low-rank bilinear    | ECFP4     | BulkFormer     | 12   | 88,705     | 0.351 (0.114) | 0.419 (0.118) | 0.051 (0.040)  | 0.0559         |
|                      | ECFP4     | CellPLM        | 12   | 84,481     | 0.493 (0.106) | 0.569 (0.126) | 0.080 (0.053)  | 0.0185         |
|                      | ECFP4     | Expression     | 12   | 594,298    | 0.266 (0.147) | 0.304 (0.170) | 0.047 (0.038)  | 0.0501         |
|                      | ECFP4     | scGPT          | 12   | 84,481     | 0.391 (0.104) | 0.466 (0.108) | 0.058 (0.050)  | 0.0431         |
|                      | KPGT      | Expression     | 4    | 602,746    | 0.181 (0.087) | 0.207 (0.100) | 0.024 (0.021)  | 0.0581         |
|                      | MolCLR    | Expression     | 4    | 543,610    | 0.074 (0.075) | 0.084 (0.085) | 0.009 (0.009)  | 0.0632         |
|                      | UniMol    | Expression     | 4    | 543,610    | 0.125 (0.092) | 0.141 (0.105) | 0.018 (0.027)  | 0.0370         |
| Nonlinear dual tower | ECFP4     | BulkFormer     | 3    | 3,812,353  | 0.845 (0.011) | 0.951 (0.009) | 0.292 (0.010)  | 0.0282         |
|                      | ECFP4     | CellPLM        | 3    | 3,681,281  | 0.856 (0.020) | 0.962 (0.015) | 0.275 (0.049)  | 0.0224         |
|                      | ECFP4     | Expression     | 3    | 19,501,057 | 0.856 (0.011) | 0.957 (0.007) | 0.332 (0.064)  | 0.0262         |
|                      | ECFP4     | scGPT          | 3    | 3,681,281  | 0.857 (0.008) | 0.962 (0.007) | 0.300 (0.028)  | 0.0257         |
|                      | KPGT      | Expression     | 1    | 19,632,129 | 0.836         | 0.933         | 0.278          | 0.0202         |
|                      | MolCLR    | Expression     | 1    | 18,714,625 | 0.817         | 0.911         | 0.288          | 0.0209         |
|                      | UniMol    | Expression     | 1    | 18,714,625 | 0.837         | 0.933         | 0.294          | 0.0219         |

**Supplementary Table 12: Agreement between representation rankings under different decoders.**  $\rho_s$  is the Spearman correlation between the two rankings named in Comparison, over the representations that both decoders were run on. Top 1 and Top 2 record whether the leading representation and the leading pair are the same under both. Molecular rows hold gene-level expression fixed and transcriptomic rows hold ECFP4 fixed. Seed 3407 rows compare the orderings of four representations under two decoders, as in Fig. 3f; three-seed rows compare a decoder with itself across three seeds and give its mean ordering. Comparisons on the interaction component are omitted where one of the two decoders emits no interaction.

| Scope     | Axis      | Component | Comparison        | $\rho_s$ | Top 1 | Top 2 | Mean ordering |
|-----------|-----------|-----------|-------------------|----------|-------|-------|---------------|
| Seed 3407 | Molecular | Total     | ridge vs bilinear | 0.80     | yes   | no    |               |
| Seed 3407 | Molecular | Total     | ridge vs tower    | 1.00     | yes   | yes   |               |

Supplementary Table 12: Continued.

| Scope       | Axis           | Component   | Comparison               | $\rho_s$ | Top 1 | Top 2 | Mean ordering                             |
|-------------|----------------|-------------|--------------------------|----------|-------|-------|-------------------------------------------|
| Seed 3407   | Molecular      | Total       | bilinear vs tower        | 0.80     | yes   | no    |                                           |
| Seed 3407   | Molecular      | Additive    | ridge vs bilinear        | 0.80     | yes   | no    |                                           |
| Seed 3407   | Molecular      | Additive    | ridge vs tower           | 1.00     | yes   | yes   |                                           |
| Seed 3407   | Molecular      | Additive    | bilinear vs tower        | 0.80     | yes   | no    |                                           |
| Seed 3407   | Molecular      | Interaction | bilinear vs tower        | 0.20     | yes   | no    |                                           |
| Seed 3407   | Transcriptomic | Total       | ridge vs bilinear        | 0.80     | no    | yes   |                                           |
| Seed 3407   | Transcriptomic | Total       | ridge vs tower           | 0.40     | yes   | no    |                                           |
| Seed 3407   | Transcriptomic | Total       | bilinear vs tower        | 0.00     | no    | no    |                                           |
| Seed 3407   | Transcriptomic | Additive    | ridge vs bilinear        | 1.00     | yes   | yes   |                                           |
| Seed 3407   | Transcriptomic | Additive    | ridge vs tower           | 0.80     | yes   | yes   |                                           |
| Seed 3407   | Transcriptomic | Additive    | bilinear vs tower        | 0.80     | yes   | yes   |                                           |
| Seed 3407   | Transcriptomic | Interaction | bilinear vs tower        | -0.80    | no    | no    |                                           |
| Three seeds | Transcriptomic | Total       | bilinear, seed stability | 0.87     | no    | —     | scGPT > CellPLM > BulkFormer > Expression |
| Three seeds | Transcriptomic | Additive    | bilinear, seed stability | 1.00     | yes   | —     | CellPLM > scGPT > BulkFormer > Expression |
| Three seeds | Transcriptomic | Interaction | bilinear, seed stability | 1.00     | yes   | —     | scGPT > CellPLM > BulkFormer > Expression |
| Three seeds | Transcriptomic | Total       | tower, seed stability    | -0.20    | no    | —     | scGPT > Expression > CellPLM > BulkFormer |
| Three seeds | Transcriptomic | Additive    | tower, seed stability    | 0.40     | no    | —     | CellPLM > scGPT > Expression > BulkFormer |
| Three seeds | Transcriptomic | Interaction | tower, seed stability    | -0.33    | no    | —     | Expression > scGPT > BulkFormer > CellPLM |

**Supplementary Table 13: Component-wise error geometry, per run.**  $d_H$  is the dimension of the additive subspace and  $d_H/N$  its share of the degrees of freedom, in per cent.  $f_H$  is the share of total squared prediction error that falls in the additive subspace.  $D_H$  and  $D_M$  are the rank-normalized error densities of the two subspaces and  $C_{\text{error}} = D_H/D_M$  their ratio.  $E_{\text{add}}$  and  $E_{\text{int}}$  are the two error terms of the exact attribution  $\text{MSE} = E_{\text{add}} + E_{\text{int}}$ . The three error quantities are given in units of  $10^{-3}$ .

| Regime              | Arm | Seed | $N$     | $d_H$  | $d_H/N$ | $f_H$ | $D_H$  | $D_M$ | $C_{\text{error}}$ | $E_{\text{add}}$ | $E_{\text{int}}$ |
|---------------------|-----|------|---------|--------|---------|-------|--------|-------|--------------------|------------------|------------------|
| Held-out cell lines | M0  | 3407 | 557,484 | 54,323 | 9.74    | 0.257 | 15.82  | 4.93  | 3.21               | 1.54             | 4.45             |
|                     | M0  | 3408 | 474,279 | 54,323 | 11.45   | 0.222 | 11.44  | 5.19  | 2.20               | 1.31             | 4.59             |
|                     | M0  | 3409 | 486,991 | 54,319 | 11.15   | 0.207 | 11.29  | 5.42  | 2.08               | 1.26             | 4.82             |
|                     | M0  | 3410 | 570,863 | 54,322 | 9.52    | 0.303 | 19.99  | 4.83  | 4.14               | 1.90             | 4.37             |
|                     | M0  | 3411 | 530,351 | 54,322 | 10.24   | 0.182 | 9.92   | 5.09  | 1.95               | 1.02             | 4.57             |
|                     | M2  | 3407 | 557,484 | 54,323 | 9.74    | 0.255 | 15.75  | 4.96  | 3.18               | 1.53             | 4.47             |
|                     | M2  | 3408 | 474,279 | 54,323 | 11.45   | 0.239 | 12.48  | 5.13  | 2.43               | 1.43             | 4.54             |
|                     | M2  | 3409 | 486,991 | 54,319 | 11.15   | 0.206 | 11.25  | 5.43  | 2.07               | 1.25             | 4.83             |
|                     | M2  | 3410 | 570,863 | 54,322 | 9.52    | 0.247 | 15.17  | 4.86  | 3.12               | 1.44             | 4.40             |
|                     | M2  | 3411 | 530,351 | 54,322 | 10.24   | 0.195 | 11.19  | 5.28  | 2.12               | 1.15             | 4.74             |
| Held-out compounds  | M0  | 3407 | 468,871 | 9,112  | 1.94    | 0.632 | 431.36 | 4.97  | 86.81              | 8.38             | 4.87             |
|                     | M0  | 3408 | 466,230 | 9,110  | 1.95    | 0.634 | 448.38 | 5.15  | 87.04              | 8.76             | 5.05             |
|                     | M0  | 3409 | 459,058 | 9,110  | 1.98    | 0.642 | 449.94 | 5.08  | 88.52              | 8.93             | 4.98             |
|                     | M0  | 3410 | 475,001 | 9,112  | 1.92    | 0.643 | 472.71 | 5.12  | 92.28              | 9.07             | 5.02             |
|                     | M0  | 3411 | 474,086 | 9,110  | 1.92    | 0.641 | 444.60 | 4.88  | 91.19              | 8.54             | 4.78             |
|                     | M2  | 3407 | 468,871 | 9,112  | 1.94    | 0.651 | 459.78 | 4.88  | 94.31              | 8.94             | 4.78             |
|                     | M2  | 3408 | 466,230 | 9,110  | 1.95    | 0.655 | 484.06 | 5.09  | 95.14              | 9.46             | 4.99             |
|                     | M2  | 3409 | 459,058 | 9,110  | 1.98    | 0.652 | 467.14 | 5.05  | 92.45              | 9.27             | 4.95             |
|                     | M2  | 3410 | 475,001 | 9,112  | 1.92    | 0.640 | 455.80 | 5.01  | 91.04              | 8.74             | 4.91             |
|                     | M2  | 3411 | 474,086 | 9,110  | 1.92    | 0.658 | 468.39 | 4.78  | 97.99              | 9.00             | 4.69             |

**Supplementary Table 14: Recovery relative to the empirical reproducibility reference, per run.**  $\eta_c = r_c / \sqrt{q_c}$  on the matched support that each regime holds out, whose size is given in the fourth column. The held-out compound rows rest on 7 to 19 held-out compounds (Supplementary Note 3).

| Regime              | Arm | Seed | Matched pairs | $\eta_{\text{total}}$ | $\eta_{\text{add}}$ | $\eta_{\text{int}}$ |
|---------------------|-----|------|---------------|-----------------------|---------------------|---------------------|
| Held-out cell lines | M0  | 3407 | 2,968         | 0.861                 | 1.032               | 0.358               |
|                     | M0  | 3408 | 3,015         | 0.829                 | 1.009               | 0.365               |
|                     | M0  | 3409 | 2,904         | 0.831                 | 1.011               | 0.361               |
|                     | M0  | 3410 | 3,051         | 0.815                 | 0.973               | 0.391               |
|                     | M0  | 3411 | 2,876         | 0.858                 | 0.997               | 0.375               |
|                     | M2  | 3407 | 2,968         | 0.833                 | 1.002               | 0.334               |
|                     | M2  | 3408 | 3,015         | 0.822                 | 1.003               | 0.375               |
|                     | M2  | 3409 | 2,904         | 0.824                 | 1.008               | 0.350               |
|                     | M2  | 3410 | 3,051         | 0.820                 | 0.980               | 0.412               |
|                     | M2  | 3411 | 2,876         | 0.855                 | 0.986               | 0.406               |
| Held-out compounds  | M0  | 3407 | 2,721         | 0.297                 | 0.322               | 0.120               |
|                     | M0  | 3408 | 3,470         | 0.365                 | 0.443               | 0.261               |
|                     | M0  | 3409 | 1,970         | 0.552                 | 0.564               | 0.538               |
|                     | M0  | 3410 | 3,873         | 0.590                 | 0.646               | 0.221               |
|                     | M0  | 3411 | 2,998         | 0.301                 | 0.338               | 0.032               |
|                     | M2  | 3407 | 2,721         | 0.344                 | 0.366               | 0.162               |
|                     | M2  | 3408 | 3,470         | 0.380                 | 0.476               | 0.239               |
|                     | M2  | 3409 | 1,970         | 0.263                 | 0.177               | 0.675               |
|                     | M2  | 3410 | 3,873         | 0.574                 | 0.630               | 0.234               |
|                     | M2  | 3411 | 2,998         | 0.140                 | 0.163               | 0.005               |

**Supplementary Table 15: Input-compatibility check, per organoid cohort.** Level and s.d. summarize the distribution of the transcriptomic input over each cohort, Gap vs CCLE is the absolute difference in level from the training reference in the first row, and Zeros is the fraction of entries equal to zero. Samples counts the expression profiles of each cohort, including organoids with no evaluated response pair; the 173 organoids of the zero-shot evaluation (Fig. 1b) are those with at least one, and Rows counts their drug–organoid pairs. *Primary* marks the primary set (Methods), and *Compatible* marks cohorts that meet all three conditions. UMPDO3 lies 0.005 outside the level threshold of 1.0 and matches UMPDO1 and UMPDO2 on zero fraction and minimum.

| Cohort                    | Samples | Genes  | Level | s.d.  | Gap vs CCLE | Zeros | Min   | Primary | Compatible | Rows  |
|---------------------------|---------|--------|-------|-------|-------------|-------|-------|---------|------------|-------|
| CCLE (training reference) | —       | —      | 2.702 | 0.221 | 0.000       | 0.153 | 0.00  | —       | —          | —     |
| HKUPDO                    | 52      | 8,866  | 0.008 | 0.075 | 2.694       | 0.951 | -9.97 | no      | no         | 324   |
| LICOB                     | 65      | 13,224 | 0.399 | 0.015 | 2.303       | 0.000 | -1.80 | no      | no         | 4,800 |
| UMPDO1                    | 90      | 15,413 | 3.578 | 0.128 | 0.876       | 0.114 | 0.00  | yes     | yes        | 2,744 |
| UMPDO2                    | 34      | 15,218 | 3.397 | 0.159 | 0.694       | 0.158 | 0.00  | yes     | yes        | 1,470 |
| UMPDO3                    | 14      | 14,651 | 3.707 | 0.198 | 1.005       | 0.094 | 0.00  | yes     | no         | 672   |

**Supplementary Table 16: Matched three-arm comparison.** M0, M1 and M2 were trained on the same dataset, data partitions and five seeds and selected by minimum validation total-response error. M1 and M2 have identical architecture and parameter count; only M2 receives component-wise supervision. Entries are mean (s.d.) over seeds.

| Regime              | Arm | Seeds | Mean pairs            | Total         | Additive      | Interaction   | $A_{\text{int}}$ | Interaction $R^2$ | Parameters |
|---------------------|-----|-------|-----------------------|---------------|---------------|---------------|------------------|-------------------|------------|
| Held-out cell lines | M0  | 5     | 523,994 0.858 (0.009) | 0.963 (0.007) | 0.315 (0.040) | 0.346 (0.042) | 0.097 (0.029)    | 19,501,057        |            |
|                     | M1  | 5     | 523,994 0.855 (0.010) | 0.960 (0.007) | 0.306 (0.044) | 0.369 (0.030) | 0.090 (0.031)    | 20,029,444        |            |
|                     | M2  | 5     | 523,994 0.858 (0.006) | 0.963 (0.004) | 0.307 (0.042) | 0.376 (0.028) | 0.090 (0.032)    | 20,029,444        |            |
| Held-out compounds  | M0  | 5     | 468,649 0.623 (0.008) | 0.697 (0.006) | 0.291 (0.017) | 0.374 (0.007) | 0.078 (0.014)    | 19,501,057        |            |
|                     | M1  | 5     | 468,649 0.618 (0.013) | 0.692 (0.012) | 0.279 (0.018) | 0.354 (0.023) | 0.072 (0.012)    | 20,029,444        |            |
|                     | M2  | 5     | 468,649 0.609 (0.013) | 0.678 (0.014) | 0.308 (0.013) | 0.359 (0.012) | 0.092 (0.009)    | 20,029,444        |            |

**Supplementary Table 17: Model choices and their independent GDSC2 endpoints.** Each criterion selects M0 or M2 on GDSC1 separately by split and seed; M2 of 5 is the number of M2 choices.  $\Delta$  MAE is the paired independent GDSC2 mean absolute error relative to global-correlation selection, so positive is worse. Sign, PCC and Spearman are secondary endpoints. The prespecified primary success rule was not met.

| Regime              | Selection criterion             | M2 of 5 | GDSC2 MAE | $\Delta$ MAE | Sign   | PCC    | Spearman |
|---------------------|---------------------------------|---------|-----------|--------------|--------|--------|----------|
| Held-out cell lines | Global                          | 1       | 0.13716   | 0.00000      | 0.5767 | 0.2815 | 0.2490   |
|                     | Fixed drug                      | 4       | 0.14079   | 0.00363      | 0.5778 | 0.2795 | 0.2489   |
|                     | Fixed cell                      | 1       | 0.13716   | 0.00000      | 0.5767 | 0.2815 | 0.2490   |
|                     | DrEval normalized               | 1       | 0.13716   | 0.00000      | 0.5767 | 0.2815 | 0.2490   |
|                     | DrugDis interaction correlation | 3       | 0.13961   | 0.00245      | 0.5792 | 0.2879 | 0.2558   |
|                     | DrugDis interaction $R^2$       | 1       | 0.13716   | 0.00000      | 0.5767 | 0.2815 | 0.2490   |
| Held-out compounds  | Global                          | 2       | 0.12980   | 0.00000      | 0.5636 | 0.2979 | 0.2269   |
|                     | Fixed drug                      | 2       | 0.13056   | 0.00076      | 0.5651 | 0.2900 | 0.2321   |
|                     | Fixed cell                      | 2       | 0.12980   | 0.00000      | 0.5636 | 0.2979 | 0.2269   |
|                     | DrEval normalized               | 2       | 0.12980   | 0.00000      | 0.5636 | 0.2979 | 0.2269   |
|                     | DrugDis interaction correlation | 4       | 0.13023   | 0.00043      | 0.5655 | 0.3094 | 0.2334   |
|                     | DrugDis interaction $R^2$       | 3       | 0.13009   | 0.00029      | 0.5662 | 0.3018 | 0.2330   |

**Supplementary Table 18: Amplitude-sensitive interaction variance explained.** Interaction is the component correlation,  $A_{\text{int}}$  is the emitted amplitude relative to the observed interaction, and  $R_{\text{int}}^2 = 2A_{\text{int}}r_{\text{int}} - A_{\text{int}}^2$ . Entries are mean (s.d.) over five runs selected by minimum validation mean squared error. Unlike a squared correlation, this quantity accounts for amplitude and can be negative.

| Regime                    | Arm | Seeds | Interaction   | $A_{\text{int}}$ | $R_{\text{int}}^2$ |
|---------------------------|-----|-------|---------------|------------------|--------------------|
| Held-out cell lines       | M0  | 5     | 0.315 (0.040) | 0.346 (0.042)    | 0.097 (0.029)      |
|                           | M2  | 5     | 0.307 (0.042) | 0.376 (0.028)    | 0.090 (0.032)      |
| Held-out compounds        | M0  | 5     | 0.291 (0.017) | 0.374 (0.007)    | 0.078 (0.014)      |
|                           | M2  | 5     | 0.308 (0.013) | 0.359 (0.012)    | 0.092 (0.009)      |
| Organoids, primary pooled | M0  | 5     | 0.067 (0.010) | 0.179 (0.049)    | -0.010 (0.012)     |
|                           | M2  | 5     | 0.087 (0.009) | 0.209 (0.021)    | -0.008 (0.007)     |

**Supplementary Table 19: Independent assay-generation selectivity evaluation by model arm.** GDSC1 is the model-selection assay generation and GDSC2 the independent test generation. Each split and seed uses 50,000 support-sampled double-difference rectangles. Values are means over five seeds.

| Regime              | Arm | Seeds | Mean pairs | GDSC1 MAE | GDSC1 sign | GDSC1 PCC | GDSC2 MAE | GDSC2 sign | GDSC2 PCC |
|---------------------|-----|-------|------------|-----------|------------|-----------|-----------|------------|-----------|
| Held-out cell lines | M0  | 5     | 5,138      | 0.1946    | 0.5696     | 0.2315    | 0.1371    | 0.5782     | 0.2817    |
|                     | M2  | 5     | 5,138      | 0.1979    | 0.5694     | 0.2252    | 0.1429    | 0.5747     | 0.2629    |
| Held-out compounds  | M0  | 5     | 5,027      | 0.1804    | 0.5525     | 0.2133    | 0.1303    | 0.5624     | 0.2837    |
|                     | M2  | 5     | 5,027      | 0.1794    | 0.5572     | 0.2361    | 0.1303    | 0.5651     | 0.3055    |

**Supplementary Table 20: The component-supervised predictor against the baseline, per seed.** M0 is the total-response-supervised baseline. M2 adds two effect heads, component-wise supervision and 2.7% trainable parameters. Arms are paired within seed, on the same split. MSE,  $E_{\text{add}}$  and  $E_{\text{int}}$  are given in units of  $10^{-3}$  and satisfy  $\text{MSE} = E_{\text{add}} + E_{\text{int}}$  exactly. Epoch is the selected checkpoint, the epoch of lowest validation total prediction error.

| Regime              | Arm | Seed | N       | Total  | Additive | Interaction | $A_{\text{int}}$ | MSE   | $E_{\text{add}}$ | $E_{\text{int}}$ | Epoch | Parameters |
|---------------------|-----|------|---------|--------|----------|-------------|------------------|-------|------------------|------------------|-------|------------|
| Held-out cell lines | M0  | 3407 | 557,484 | 0.8671 | 0.9661   | 0.3160      | 0.346            | 5.99  | 1.54             | 4.45             | 16    | 19,501,057 |
|                     | M0  | 3408 | 474,279 | 0.8566 | 0.9608   | 0.3561      | 0.356            | 5.90  | 1.31             | 4.59             | 24    | 19,501,057 |
|                     | M0  | 3409 | 486,991 | 0.8535 | 0.9642   | 0.2507      | 0.365            | 6.07  | 1.26             | 4.82             | 30    | 19,501,057 |
|                     | M0  | 3410 | 570,863 | 0.8466 | 0.9514   | 0.3173      | 0.277            | 6.27  | 1.90             | 4.37             | 12    | 19,501,057 |
|                     | M0  | 3411 | 530,351 | 0.8667 | 0.9714   | 0.3371      | 0.388            | 5.59  | 1.02             | 4.57             | 30    | 19,501,057 |
|                     | M2  | 3407 | 557,484 | 0.8653 | 0.9654   | 0.3101      | 0.364            | 6.01  | 1.53             | 4.47             | 19    | 20,029,444 |
|                     | M2  | 3408 | 474,279 | 0.8604 | 0.9640   | 0.3724      | 0.416            | 5.97  | 1.43             | 4.54             | 24    | 20,029,444 |
|                     | M2  | 3409 | 486,991 | 0.8547 | 0.9648   | 0.2570      | 0.394            | 6.08  | 1.25             | 4.83             | 27    | 20,029,444 |
|                     | M2  | 3410 | 570,863 | 0.8501 | 0.9556   | 0.3087      | 0.357            | 5.84  | 1.44             | 4.40             | 20    | 20,029,444 |
|                     | M2  | 3411 | 530,351 | 0.8588 | 0.9668   | 0.2886      | 0.351            | 5.88  | 1.15             | 4.74             | 19    | 20,029,444 |
| Held-out compounds  | M0  | 3407 | 468,871 | 0.6157 | 0.6925   | 0.2869      | 0.373            | 13.26 | 8.38             | 4.87             | 22    | 19,501,057 |

**Supplementary Table 20: Continued.**

| Regime | Arm | Seed | <i>N</i> | Total  | Additive | Interaction | $A_{\text{int}}$ | MSE   | $E_{\text{add}}$ | $E_{\text{int}}$ | Epoch | Parameters |
|--------|-----|------|----------|--------|----------|-------------|------------------|-------|------------------|------------------|-------|------------|
|        | M0  | 3408 | 466,230  | 0.6358 | 0.7083   | 0.3154      | 0.369            | 13.81 | 8.76             | 5.05             | 25    | 19,501,057 |
|        | M0  | 3409 | 459,058  | 0.6244 | 0.6965   | 0.3012      | 0.370            | 13.91 | 8.93             | 4.98             | 22    | 19,501,057 |
|        | M0  | 3410 | 475,001  | 0.6201 | 0.6953   | 0.2750      | 0.372            | 14.09 | 9.07             | 5.02             | 14    | 19,501,057 |
|        | M0  | 3411 | 474,086  | 0.6195 | 0.6940   | 0.2762      | 0.386            | 13.33 | 8.54             | 4.78             | 26    | 19,501,057 |
|        | M2  | 3407 | 468,871  | 0.5900 | 0.6584   | 0.3086      | 0.362            | 13.72 | 8.94             | 4.78             | 22    | 20,029,444 |
|        | M2  | 3408 | 466,230  | 0.6144 | 0.6806   | 0.3290      | 0.351            | 14.45 | 9.46             | 4.99             | 21    | 20,029,444 |
|        | M2  | 3409 | 459,058  | 0.6126 | 0.6817   | 0.3051      | 0.346            | 14.22 | 9.27             | 4.95             | 16    | 20,029,444 |
|        | M2  | 3410 | 475,001  | 0.6233 | 0.6960   | 0.3044      | 0.376            | 13.65 | 8.74             | 4.91             | 28    | 20,029,444 |
|        | M2  | 3411 | 474,086  | 0.6034 | 0.6723   | 0.2947      | 0.361            | 13.69 | 9.00             | 4.69             | 30    | 20,029,444 |

**Supplementary Table 21: Zero-shot organoid evaluation, pooled and per cohort.** Mean and standard deviation across five seeds. The first block is the primary analysis over UMPDO1–3, of which UMPDO1–2 meet every compatibility condition and UMPDO3 misses the level condition by 0.005; the second pools all five cohorts; the remaining rows give each cohort separately, with its Primary and Compatible flags.

| Cohort              | Primary | Compatible | Arm | Seeds | <i>N</i> | Total         | Additive      | Interaction    | $A_{\text{int}}$ |
|---------------------|---------|------------|-----|-------|----------|---------------|---------------|----------------|------------------|
| Primary pooled      |         |            | M0  | 5     | 4,886    | 0.460 (0.007) | 0.607 (0.011) | 0.067 (0.010)  | 0.179 (0.049)    |
|                     |         |            | M2  | 5     | 4,886    | 0.466 (0.036) | 0.614 (0.046) | 0.087 (0.009)  | 0.209 (0.021)    |
| All cohorts, pooled |         |            | M0  | 5     | 10,010   | 0.271 (0.097) | 0.356 (0.128) | 0.040 (0.030)  | 0.206 (0.041)    |
|                     |         |            | M2  | 5     | 10,010   | 0.382 (0.048) | 0.500 (0.058) | 0.078 (0.026)  | 0.235 (0.015)    |
| HKUPDO              | no      | no         | M0  | 5     | 324      | 0.442 (0.088) | 0.493 (0.106) | -0.005 (0.092) | 0.563 (0.165)    |
|                     | no      | no         | M2  | 5     | 324      | 0.358 (0.096) | 0.402 (0.106) | -0.015 (0.096) | 0.736 (0.051)    |
| LICOB               | no      | no         | M0  | 5     | 4,800    | 0.268 (0.045) | 0.377 (0.063) | 0.066 (0.019)  | 0.118 (0.096)    |
|                     | no      | no         | M2  | 5     | 4,800    | 0.283 (0.035) | 0.402 (0.036) | 0.054 (0.066)  | 0.224 (0.075)    |
| UMPDO1              | yes     | yes        | M0  | 5     | 2,744    | 0.561 (0.014) | 0.662 (0.015) | 0.054 (0.017)  | 0.221 (0.049)    |
|                     | yes     | yes        | M2  | 5     | 2,744    | 0.547 (0.034) | 0.648 (0.036) | 0.077 (0.013)  | 0.279 (0.023)    |
| UMPDO2              | yes     | yes        | M0  | 5     | 1,470    | 0.358 (0.011) | 0.552 (0.017) | 0.058 (0.019)  | 0.092 (0.017)    |
|                     | yes     | yes        | M2  | 5     | 1,470    | 0.364 (0.023) | 0.560 (0.037) | 0.069 (0.014)  | 0.097 (0.024)    |
| UMPDO3              | yes     | no         | M0  | 5     | 672      | 0.383 (0.024) | 0.461 (0.038) | 0.140 (0.052)  | 0.147 (0.036)    |
|                     | yes     | no         | M2  | 5     | 672      | 0.391 (0.048) | 0.466 (0.049) | 0.167 (0.107)  | 0.187 (0.009)    |

**Supplementary Table 22: Zero-shot organoid evaluation of the matched control M1.** Held-out-cell-line checkpoints were applied to the organoids. The primary pooled set is UMPDO1, UMPDO2 and UMPDO3, and M0 and M2 are shown for the three-arm comparison. Entries are mean (s.d.) over five seeds.

| Cohort              | Arm | Seeds | <i>N</i> | Additive      | Interaction    | $A_{\text{int}}$ | Interaction $R^2$ |
|---------------------|-----|-------|----------|---------------|----------------|------------------|-------------------|
| Primary pooled      | M0  | 5     | 4,886    | 0.607 (0.011) | 0.067 (0.010)  | 0.179 (0.049)    | -0.010 (0.012)    |
|                     | M1  | 5     | 4,886    | 0.621 (0.024) | 0.061 (0.013)  | 0.165 (0.059)    | -0.009 (0.011)    |
|                     | M2  | 5     | 4,886    | 0.614 (0.046) | 0.087 (0.009)  | 0.209 (0.021)    | -0.008 (0.007)    |
| UMPDO1              | M0  | 5     | 2,744    | 0.662 (0.015) | 0.054 (0.017)  | 0.221 (0.049)    | -0.028 (0.020)    |
|                     | M1  | 5     | 2,744    | 0.653 (0.028) | 0.048 (0.026)  | 0.213 (0.099)    | -0.031 (0.032)    |
|                     | M2  | 5     | 2,744    | 0.648 (0.036) | 0.077 (0.013)  | 0.279 (0.023)    | -0.035 (0.010)    |
| UMPDO2              | M0  | 5     | 1,470    | 0.552 (0.017) | 0.058 (0.019)  | 0.092 (0.017)    | 0.002 (0.003)     |
|                     | M1  | 5     | 1,470    | 0.548 (0.025) | 0.052 (0.018)  | 0.074 (0.020)    | 0.002 (0.004)     |
|                     | M2  | 5     | 1,470    | 0.560 (0.037) | 0.069 (0.014)  | 0.097 (0.024)    | 0.004 (0.002)     |
| UMPDO3              | M0  | 5     | 672      | 0.461 (0.038) | 0.140 (0.052)  | 0.147 (0.036)    | 0.019 (0.013)     |
|                     | M1  | 5     | 672      | 0.483 (0.051) | 0.115 (0.089)  | 0.136 (0.059)    | 0.015 (0.017)     |
|                     | M2  | 5     | 672      | 0.466 (0.049) | 0.167 (0.107)  | 0.187 (0.009)    | 0.028 (0.040)     |
| All cohorts, pooled | M0  | 5     | 10,010   | 0.356 (0.128) | 0.040 (0.030)  | 0.206 (0.041)    | -0.027 (0.016)    |
|                     | M1  | 5     | 10,010   | 0.354 (0.150) | 0.032 (0.031)  | 0.186 (0.038)    | -0.023 (0.009)    |
|                     | M2  | 5     | 10,010   | 0.500 (0.058) | 0.078 (0.026)  | 0.235 (0.015)    | -0.019 (0.011)    |
| LICOB               | M0  | 5     | 4,800    | 0.377 (0.063) | 0.066 (0.019)  | 0.118 (0.096)    | -0.005 (0.021)    |
|                     | M1  | 5     | 4,800    | 0.410 (0.026) | 0.072 (0.048)  | 0.106 (0.059)    | 0.002 (0.013)     |
|                     | M2  | 5     | 4,800    | 0.402 (0.036) | 0.054 (0.066)  | 0.224 (0.075)    | -0.032 (0.047)    |
| HKUPDO              | M0  | 5     | 324      | 0.493 (0.106) | -0.005 (0.092) | 0.563 (0.165)    | -0.326 (0.154)    |
|                     | M1  | 5     | 324      | 0.468 (0.096) | 0.013 (0.054)  | 0.690 (0.166)    | -0.482 (0.264)    |
|                     | M2  | 5     | 324      | 0.402 (0.106) | -0.015 (0.096) | 0.736 (0.051)    | -0.565 (0.146)    |

## Supplementary Data

### Supplementary Data 1: Data sources

Supplementary Data 1 lists each response resource and organoid cohort used in this study, with its use, the DROMA tables read, the rows used and its primary publication. All data were read from the processed DROMA collection ([10.5281/zenodo.18503188](https://doi.org/10.5281/zenodo.18503188)). Rows are response rows in the benchmark dataset for a resource and rows scored in the zero-shot cross-system evaluation for a cohort.

| Resource or cohort | Use in this study                                                                    | Rows      | Primary publication                           |
|--------------------|--------------------------------------------------------------------------------------|-----------|-----------------------------------------------|
| NCI60              | Benchmark dataset                                                                    | 2,175,581 | Holbeck et al., Mol. Cancer Ther. (2010)      |
| PRISM              | Benchmark dataset                                                                    | 504,493   | Corsello et al., Nat. Cancer (2020)           |
| CTRP2              | Benchmark dataset                                                                    | 269,725   | Seashore-Ludlow et al., Cancer Discov. (2015) |
| GDSC1              | Benchmark dataset; cross-assay reproducibility; model-selection test (selection arm) | 125,914   | Garnett et al., Nature (2012)                 |
| GDSC2              | Benchmark dataset; cross-assay reproducibility; model-selection test (test arm)      | 44,797    | Iorio et al., Cell (2016)                     |
| CTRP1              | Benchmark dataset                                                                    | 39,784    | Basu et al., Cell (2013)                      |
| CCLE               | Benchmark dataset; transcriptomic input matrix                                       | 10,730    | Barretina et al., Nature (2012)               |
| GRAY               | Benchmark dataset                                                                    | 1,888     | Daemen et al., Genome Biol. (2013)            |
| gCSI               | Benchmark dataset                                                                    | 1,646     | Haverty et al., Nature (2016)                 |
| FIMM               | Benchmark dataset                                                                    | 890       | Pemovska et al., Cancer Discov. (2013)        |
| UHNBreast          | Benchmark dataset                                                                    | 54        | Marcotte et al., Cell (2016)                  |
| UMPDO1             | Zero-shot evaluation, primary set                                                    | 2,744     | —                                             |
| UMPDO2             | Zero-shot evaluation, primary set                                                    | 1,470     | —                                             |
| UMPDO3             | Zero-shot evaluation, primary set                                                    | 672       | —                                             |
| LICOB              | Zero-shot evaluation, input-incompatible                                             | 4,800     | Ji et al., Sci. Transl. Med. (2023)           |
| HKUPDO             | Zero-shot evaluation, input-incompatible                                             | 324       | —                                             |